

where\*\*

$$\begin{aligned}
 K^\mu = & \frac{1}{2} i \sum_A \epsilon^{\rho\sigma\mu\nu} f_{A\rho\sigma} \gamma_\nu \gamma_5 \lambda_A + \frac{1}{8} \sum_A [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A f_{A\rho\sigma} + \frac{1}{2} i \sum_A D_A \gamma_5 \gamma^\mu \lambda_A \\
 & - i \sum_{Anm} (t_A)_{nm} \gamma_5 \gamma^\mu \lambda_A \phi_n^* \phi_m \\
 & + \frac{1}{\sqrt{2}} \sum_n \gamma^\mu \left[ -(\not{D}\phi)_n \psi_{nR} - (\not{D}\phi)_n^* \psi_{nL} + \mathcal{F}_n^* \psi_{nL} + \mathcal{F}_n \psi_{nR} \right. \\
 & \left. + 2 \left( \frac{\partial f(\phi)}{\partial \phi_n} \right) \psi_{nL} + 2 \left( \frac{\partial f(\phi)}{\partial \phi_n} \right)^* \psi_{nR} \right]. \quad (27.4.39)
 \end{aligned}$$

The first two terms are derived using the identity (27.2.5). Using the same identity again with Eq. (26.7.4) gives the total supersymmetry current:

$$\begin{aligned}
 S^\mu = & N^\mu + K^\mu \\
 = & -\frac{1}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A - i \sum_{Anm} (t_A)_{nm} \gamma_5 \gamma^\mu \lambda_A \phi_n^* \phi_m \\
 & + \frac{1}{\sqrt{2}} \sum_n \left[ (\not{D}\phi)_n \gamma^\mu \psi_{nR} + (\not{D}\phi)_n^* \gamma^\mu \psi_{nL} \right. \\
 & \left. + 2 \left( \frac{\partial f(\phi)}{\partial \phi_n} \right) \gamma^\mu \psi_{nL} + 2 \left( \frac{\partial f(\phi)}{\partial \phi_n} \right)^* \gamma^\mu \psi_{nR} \right]. \quad (27.4.40)
 \end{aligned}$$

\* \* \*

In Section 26.8 we considered a class of supersymmetric theories with a superpotential  $f(\Phi)$  that has an arbitrary dependence on a set of left-chiral scalar superfields  $\Phi_n$  but not their derivatives, and with a Kahler potential  $K(\Phi, \Phi^*)$  that has an arbitrary dependence on the  $\Phi_n$  and  $\Phi_n^*$ , but not their derivatives. We can extend the same considerations to gauge theories, again with the dependence of the Lagrangian on the chiral superfields limited only by supersymmetry, but without introducing new superderivatives or spacetime derivatives. The renormalizable Lagrangian density is then replaced with

$$\begin{aligned}
 \mathcal{L} = & \frac{1}{2} \left[ K(\Phi, \Phi^\dagger \exp(-2 \sum_A t_A V_A)) \right]_D + 2 \operatorname{Re} [f(\Phi)]_{\mathcal{F}} \\
 & - \frac{1}{2} \operatorname{Re} \sum_{AB} \left[ h_{AB}(\Phi) (W_{AL}^T \epsilon W_{BL}) \right]_{\mathcal{F}}, \quad (27.4.41)
 \end{aligned}$$

\*\* The easiest way to calculate the change in the term  $[\Phi^\dagger \exp(-2 \sum_A t_A V_A) \Phi]_D$  is to calculate the  $\lambda$ -component of  $\Phi^\dagger \exp(-2 \sum_A t_A V_A) \Phi$  and use Eq. (26.2.17). In this way of doing the calculation, the important term on the second line of the right-hand side of Eq. (27.4.39) arises from the  $\lambda$ -component of  $\exp(-2 \sum_A t_A V_A)$ .